

Nov 3, 2024 (Due: 08:00 Nov 10, 2025)

1. Let $(\hat{\lambda}, \hat{x})$ be an approximate eigenpair of a Hermitian matrix A , and $r = A\hat{x} - \hat{x}\hat{\lambda}$ be the residual. Suppose that $\|\hat{x}\|_2 = 1$. Show that there exists an eigenvalue λ of A such that

$$|\lambda - \hat{\lambda}| \leq \|r\|_2.$$

2. Given $x, y \in \mathbb{R}^n$. Describe in detail how to construct a rotation matrix Q such that the columns of $[x, y]Q$ are orthogonal to each other.

3. Let $D \in \mathbb{R}^{n \times n}$ be diagonal with distinct eigenvalues, $z \in \mathbb{R}^n$ be a vector with no zero entries, and $\rho \in \mathbb{R} \setminus \{0\}$. Suppose that (λ, u) is an eigenpair of $D + \rho z z^\top$. Show that

- (1) $\lambda I - D$ is nonsingular;
- (2) $(\lambda I - D)^{-1}z$ is an eigenvector of $D + \rho z z^\top$;
- (3) $z^\top u \neq 0$.

4. Randomly generate a relatively small (e.g., 6×6) real symmetric matrix of the form $A = \text{diag}\{d_1, d_2, \dots, d_n\} + z z^\top$, where d_i 's are distinct, and z does not have any zero entry. Visualize the function

$$f(\lambda) = 1 - \sum_{i=1}^n \frac{z_i^2}{\lambda - d_i}.$$

Highlight the eigenvalues of A in the plot and make sure they match the roots of $f(\lambda)$. For simplicity, you may compute the eigenvalues of A by existing functions from math libraries (e.g., `eig` from MATLAB/Octave).

5. Implement the Jacobi diagonalization algorithm for real symmetric matrices. Visualize the performance for a few matrices with different sizes. Visualize the convergence history for one example. You are encouraged to make observations on *componentwise* convergence.

You are also encouraged to try the “wrong” choice of Jacobi rotations for the cyclic Jacobi algorithm.

6. (optional) Let $D = \text{diag}\{d_1, \dots, d_n\}$ be a real diagonal matrix. Let $\alpha_1, \dots, \alpha_n$ be real scalars that satisfy

$$d_n < \alpha_n < \dots < d_i < \alpha_i < d_{i-1} < \alpha_{i-1} < \dots < d_1 < \alpha_1.$$

Show that the α_i 's are the exact eigenvalues of $D + uu^\top$, where entries of the real vector u are defined by

$$u_i = \left(\frac{\prod_{1 \leq j \leq n} (\alpha_j - d_i)}{\prod_{1 \leq j \leq n, j \neq i} (d_j - d_i)} \right)^{1/2}, \quad (1 \leq i \leq n).$$

7. (optional) When bidiagonalizing an $m \times n$ matrix with $m > n$, there are two common options: bidiagonalization after QR factorization vs. direct bidiagonalization. Suppose that both the left and right orthogonal transformations need to be accumulated. Calculate the cost in terms of number of floating-point operations for these options, and identify the crossover point.

Would the crossover point change if the orthogonal transformations were not accumulated?